

Bovogo Pricing Engine

Cost-share calculation, static pricing, and revenue architecture

Bovogo / WeGotcha LLC • K. Sushant, Founder & CEO • Internal Engineering Spec v1.0 • 2026

1 OVERVIEW

This document specifies the complete pricing engine for Bovogo — the cost-share calculation, static pricing model, platform fee structure, insurance attachment, luggage surcharges, and revenue math. It is the authoritative reference for backend engineers implementing the pricing module and for anyone modelling unit economics.

All design decisions in this document were made to satisfy three simultaneous constraints: (1) maintain cost-sharing legal characterisation to avoid TNC classification, (2) give riders a predictable, static price regardless of seat occupancy, and (3) generate sustainable platform revenue through fees and insurance commissions.

Legal constraint — read before touching pricing logic

The base cost-share (what the driver receives) must never exceed the IRS standard mileage rate applied to the driver's share of the trip. The platform cannot profit from the cost-share portion. Platform revenue comes exclusively from the platform fee and insurance commission — separate line items collected on top of cost-share, never embedded within it.

2 KEY CONSTANTS AND CONFIGURATION

All pricing constants are stored as a single configuration object in the codebase. They must never be hard-coded inline. Admin tooling must allow the IRS rate to be updated each January without a code deployment.

Constant	Value	Description	Update cadence
IRS_RATE	\$0.70 / mile	2025 IRS standard mileage rate. Use as ceiling.	Every January
SAFETY_FACTOR	0.75	75% of IRS rate — cushion below the legal ceiling.	Rarely (legal review)
MIN_DISTANCE_MILES	60 miles	Trips shorter than this are rejected at posting.	Product decision
MAX_DISTANCE_MILES	400 miles	Soft cap; longer trips require ops review.	Product decision
PLATFORM_FEE	\$2.50	Flat per-booking fee. Rider-facing as 'service fee'.	Quarterly review
INSURANCE_PREMIUM	\$9.00	Per-trip insurance add-on. Default on at checkout.	MGA contract

INSURANCE_COMMISSION	0.25%	Platform commission on insurance premium.	MGA contract
SEDAN_OCCUPANCY	3	Driver + 2 assumed riders. Pricing denominator.	Product decision
SUV_OCCUPANCY	4	Driver + 3 assumed riders. Pricing denominator.	Product decision
MINIVAN_OCCUPANCY	4	Same as SUV.	Product decision

Why 0.75 and not 1.0?

Using the full IRS rate as the cost-share price creates a defensibility risk: the IRS rate includes depreciation, a non-cash cost the driver was incurring regardless of the trip. Setting the multiplier at 0.75 keeps us safely below the ceiling and more accurately reflects actual variable trip costs (fuel + incremental wear). Legal counsel has reviewed this approach for Texas. Re-confirm if expanding to new states.

3 STATIC PRICING MODEL

3.1 The Problem with Dynamic Pricing

A naive implementation would divide total trip cost among the actual number of booked riders. This creates two severe problems:

- Price changes as riders book — the first rider sees \$67.50, the third sees \$22.50. Riders who book early feel cheated when latecomers get a discount.
- The driver's recovery depends on fill rate, creating uncertainty that disincentivises posting trips on low-demand days.

3.2 The Standard Occupancy Solution

Bovogo uses a fixed standard occupancy denominator per vehicle class. This is the assumed number of people sharing the trip for pricing purposes — not the actual booking count. The price shown to every rider is calculated at this standard occupancy and never changes, regardless of how many seats ultimately fill.

Vehicle class	Total seats	Max rider seats	Standard occupancy (pricing denominator)	Rationale
Sedan	4	3	3 (driver + 2)	Expected fill 1.8–2.2 on popular corridors
SUV	6	5	4 (driver + 3)	Prevents price floor collapse vs. sedan
Minivan	7	6	4 (driver + 3)	Same floor as SUV — operational parity

Driver risk: if a sedan driver posts 3 seats and only 1 books, they collect $1 \times \$31.50$ rather than $3 \times \$31.50$. They bear the empty-seat risk. This is intentional — it incentivises drivers to post trips on high-demand corridors/times and self-regulate supply.

4 CORE PRICING FORMULA

4.1 Base Seat Price (Driver's Cost-Share Recovery)

The base seat price is what the rider pays toward the driver's costs:

```
base_seat_price = (distance_miles x IRS_RATE x SAFETY_FACTOR) / STANDARD_OCCUPANCY

# Sedan example – 180 miles:
base_seat_price = (180 x 0.70 x 0.75) / 3 = $31.50

# SUV example – 180 miles:
base_seat_price = (180 x 0.70 x 0.75) / 4 = $23.63
```

4.2 Total Rider Charge

The rider pays three components on top of base seat price:

```
rider_total = base_seat_price
+ PLATFORM_FEE # $2.50 (always)
+ luggage_surcharge # $0 / $3 / $6 / $8
+ insurance_premium # $9.00 (default on)

# Sedan, 180 miles, carry-on only, insurance elected:
rider_total = $31.50 + $2.50 + $0 + $9.00 = $43.00

# SUV, 180 miles, carry-on only, insurance elected:
rider_total = $23.63 + $2.50 + $0 + $9.00 = $35.13
```

4.3 Driver Payout

The driver receives the base seat price multiplied by the number of seats *actually booked*, plus any luggage surcharges. The driver never receives the platform fee or insurance premium — those go to the platform and MGA respectively.

```
driver_payout = (base_seat_price x seats_booked) + sum(luggage_surcharges)

# Sedan, 180 miles, 2 seats booked, both carry-on only:
driver_payout = ($31.50 x 2) + $0 = $63.00

# Sedan, 180 miles, 1 seat booked, 2 large bags:
driver_payout = ($31.50 x 1) + $3 = $34.50
```

4.4 Platform Revenue Per Rider

```
gross_platform_revenue = PLATFORM_FEE + (insurance_premium x INSURANCE_COMMISSION)
= $2.50 + ($9.00 x 0.25)
= $2.50 + $2.25
```

```

= $4.75 per rider (gross)

# Stripe processing deduction (~$0.37 on the $2.50 platform fee):
net_platform_revenue = ~$4.38 per rider

# Note: insurance commission is paid by MGA — no Stripe processing on it.
# Stripe applies only to the $2.50 platform fee portion.

```

Platform revenue is constant across vehicle types

Platform revenue per rider is always \$4.75 gross (\$4.38 net) regardless of whether the rider is in a sedan or SUV. This simplifies financial modelling and means our revenue doesn't depend on the driver's vehicle choice.

5 WORKED EXAMPLE — 180-MILE TRIP (AUSTIN TO HOUSTON)

5.1 Full Pricing Stack — Sedan

Component	Calculation	Amount	Goes to
Base cost-share	$180 \times \$0.70 \times 0.75 \div 3$	\$31.50	Driver
Platform fee	Flat fee	\$2.50	Platform
Insurance premium	Add-on (default on)	\$9.00	MGA (insurer)
Platform insurance commission	$\$9.00 \times 25\%$	\$2.25	Platform (from MGA)
RIDER PAYS TOTAL		\$43.00	
PLATFORM REVENUE (GROSS)	Fee + commission	\$4.75	Platform

5.2 Full Pricing Stack — SUV

Component	Calculation	Amount	Goes to
Base cost-share	$180 \times \$0.70 \times 0.75 \div 4$	\$23.63	Driver
Platform fee	Flat fee	\$2.50	Platform
Insurance premium	Add-on (default on)	\$9.00	MGA (insurer)
Platform insurance commission	$\$9.00 \times 25\%$	\$2.25	Platform (from MGA)
RIDER PAYS TOTAL		\$35.13	
PLATFORM REVENUE (GROSS)	Fee + commission	\$4.75	Platform

6 FULL PRICING REFERENCE TABLE

Static prices for common route distances. Insurance elected (default on). No luggage surcharge. Platform revenue is constant at \$4.75 per rider regardless of distance or vehicle.

6.1 Sedan (standard occupancy: 3)

Distance	Base cost-share	Platform fee	Insurance	Rider total	Platform rev.
60 miles	\$10.50	\$2.50	\$9.00	\$22.00	\$4.75
80 miles	\$14.00	\$2.50	\$9.00	\$25.50	\$4.75
100 miles	\$17.50	\$2.50	\$9.00	\$29.00	\$4.75
120 miles	\$21.00	\$2.50	\$9.00	\$32.50	\$4.75
150 miles	\$26.25	\$2.50	\$9.00	\$37.75	\$4.75
180 miles	\$31.50	\$2.50	\$9.00	\$43.00	\$4.75
200 miles	\$35.00	\$2.50	\$9.00	\$46.50	\$4.75
250 miles	\$43.75	\$2.50	\$9.00	\$55.25	\$4.75
300 miles	\$52.50	\$2.50	\$9.00	\$64.00	\$4.75
400 miles	\$70.00	\$2.50	\$9.00	\$81.50	\$4.75

180-mile row (Austin↔Houston) highlighted in teal.

6.2 SUV (standard occupancy: 4)

Distance	Base cost-share	Platform fee	Insurance	Rider total	Platform rev.
60 miles	\$7.88	\$2.50	\$9.00	\$19.38	\$4.75
80 miles	\$10.50	\$2.50	\$9.00	\$22.00	\$4.75
100 miles	\$13.12	\$2.50	\$9.00	\$24.62	\$4.75
120 miles	\$15.75	\$2.50	\$9.00	\$27.25	\$4.75
150 miles	\$19.69	\$2.50	\$9.00	\$31.19	\$4.75
180 miles	\$23.62	\$2.50	\$9.00	\$35.12	\$4.75
200 miles	\$26.25	\$2.50	\$9.00	\$37.75	\$4.75
250 miles	\$32.81	\$2.50	\$9.00	\$44.31	\$4.75
300 miles	\$39.38	\$2.50	\$9.00	\$50.88	\$4.75
400 miles	\$52.50	\$2.50	\$9.00	\$64.00	\$4.75

7 LUGGAGE SURCHARGE SYSTEM

Luggage surcharges compensate the driver for increased fuel cost from extra weight and for using available trunk capacity. They pass through entirely to the driver — the platform does not retain the surcharge itself. Platform revenue on luggage comes exclusively from optional luggage insurance (a separate add-on).

Tier	Luggage declared	Rider surcharge (cost recovery)	Goes to driver	Optional luggage insurance	Platform commission
carry_on	Carry-on bags only	\$0	\$0	\$0 (covered by base)	\$0
standard	1–2 large suitcases	+\$3.00	\$3.00	\$3.00 add-on	~\$0.60
large	3–4 large suitcases	+\$6.00	\$6.00	\$5.00 add-on	~\$1.00
oversized	Oversized / instruments	+\$8.00	\$8.00	\$8.00 add-on	~\$1.30

Luggage example: Sedan, 180 miles, rider with 2 large suitcases, insurance on

```

base_seat_price = $31.50 (standard sedan pricing)
platform_fee = $2.50
luggage_surcharge = $3.00 (standard tier, goes to driver)
insurance = $9.00 (base trip insurance, default on)
luggage_insurance = $3.00 (optional luggage add-on, elected)
-----
rider_total = $49.00

driver_receives = $31.50 + $3.00 = $34.50
MGA receives = $9.00 + $3.00 = $12.00 (insurance premiums)
platform_revenue = $2.50 + $2.25 + $0.60 = $5.35 (gross)

```

8 INSURANCE — DEFAULT-ON BEHAVIOUR

Trip insurance is **default on** at checkout. The rider must actively uncheck it to remove it. This is standard practice in travel e-commerce (Booking.com, Expedia, OTAs universally use this pattern). It is legally permissible provided the disclosure is clear and removal is unambiguous.

Checkout UI requirements (for frontend engineers)

- Insurance line item must be clearly visible in the order summary before payment capture.
- Label: "Trip insurance — \$9.00" with a brief description: "Covers accidental injury and luggage loss during your trip."
- A clearly visible "Remove" link or toggle must be adjacent. Tap-to-remove, no confirmation modal required.

- If removed, a single interstitial: "Are you sure? You'll be travelling without accident coverage." One-tap dismiss.
- Removed state is stored on the booking record. Default always resets to ON for next booking.
- Analytics event fired on every removal (attach-rate monitoring).

Metric	Target (Year 1)	Target (Year 2)
Insurance attach rate	50%+	60%+
Platform commission per attached rider	\$2.25	\$2.25
Expected blended insurance rev / rider	\$1.13	\$1.35

9 MONEY FLOW — STRIPE CONNECT

All payments processed via Stripe Connect Express. The platform acts as merchant of record. On booking confirmation, funds are captured and held until trip completion, then split via Stripe transfers.

```
# On booking confirmation – sedan, 180mi, insurance on, 1 rider:

Total charged to rider = $43.00

# On trip completion (T + 0 to 24h payout):

Stripe platform fee = (2.9% x $43.00) + $0.30 = $1.55

Net after Stripe = $43.00 - $1.55 = $41.45

Transfer to driver = $31.50 (base cost-share)

Transfer to MGA = $6.75 (insurance net of commission)

Platform retains = $41.45 - $31.50 - $6.75 = $3.20

# Platform gross revenue: $4.75 (fee + commission)

# Stripe cost: $1.55

# Platform net: $3.20

# Note: MGA transfer is handled via separate API call to MGA partner,
# not as a Stripe sub-transfer. Platform receives full insurance premium
# from rider, then remits net premium to MGA and retains commission.
```

10 PSEUDOCODE — COMPLETE IMPLEMENTATION REFERENCE

[illegible]

[illegible]

```

platform_fees = PLATFORM_FEE * COUNT(bookings)
insurance_commiss = INSURANCE_PREMIUM * INSURANCE_COMMISSION * insured_count
gross = platform_fees + insurance_commiss
stripe_cost = (0.029 * gross) + (0.30 * COUNT(bookings))
RETURN {
  gross: gross,
  net: gross - stripe_cost,
  breakdown: { platform_fees, insurance_commiss, stripe_cost }
}

```

11 UNIT ECONOMICS SUMMARY

Based on the pricing model above, at mature Year-2 cohort assumptions (50% insurance attach, 11% premium subscription conversion):

Metric	Value	Notes
Avg. revenue per trip (2 riders avg)	\$4.75 gross	Insurance commission + platform fees
Variable cost per trip	\$2.20	Stripe, KYC amort., SMS, T&S allocation
Contribution margin per trip	\$2.55	54% margin
Blended customer acquisition cost	\$25.00	Weighted driver + rider
LTV — 2-year cohort (15 trips/yr)	~\$77	Based on repeat rate assumptions
LTV : CAC ratio	3.1 : 1	Healthy for early-stage marketplace
Year 5 projected ARR (multi-state)	\$18M	Conservative; excludes affiliate revenue

How to improve these numbers over time

Three levers: (1) Increase insurance attach rate beyond 50% through better checkout UX and social proof — each +10% attach adds ~\$0.23 gross revenue per trip. (2) Negotiate MGA commission upward from 25% as volume grows — 30-35% is achievable at Series A scale, adding ~\$0.45 per attached rider. (3) Reduce CAC through organic growth, referral programmes, and campus ambassador networks — a \$5 reduction in CAC improves LTV:CAC from 3.1 to 3.5.

12 EDGE CASES AND GUARD RAILS

Scenario	Behaviour	Rationale
Trip < 60 miles	Reject at posting. Show: 'Bovogo serves intercity trips of 60+ miles.'	Below 60mi blurs TNC line; value prop collapses

Trip > 400 miles	Soft cap — allow with ops review flag. Do not auto-reject.	Driver fatigue risk; edge of use-case
IRS rate updated	Admin updates constant in config. All future trips use new rate. In-flight trips keep locked price.	Rate is locked at trip-post time
Driver cancels trip	Full refund to all riders. Driver penalty applied. No platform fee retention.	Driver breach; platform not at fault
Rider cancels > 24h out	Full refund minus Stripe processing. Platform fee refunded.	Standard consumer protection
Rider cancels < 6h out	50% refund. Platform fee retained.	Late cancel compensates driver for lost fill
0 riders book	Driver can cancel penalty-free up to 4h before departure.	No reason to penalise unfilled posting
Insurance declined by rider	Remove \$9.00 from total. Log as insurance_elected = false. Analytics event fired.	Rider's right; log for attach-rate tracking
Vehicle type not declared	Default to sedan pricing (most conservative — higher price, safer floor).	Fail safe toward driver; rider can query before booking

13 OPEN DECISIONS FOR TECHNICAL CO-FOUNDER

The following decisions are not yet made and should be resolved during the first engineering sprint:

- Should pricing be displayed to the rider as a breakdown (base + fee + insurance) or as a single total?
Recommendation: single total at search, breakdown at checkout confirmation.
- Should the driver see a projected earnings estimate when posting (e.g. 'You could earn up to \$63 if all seats fill')? Risk: may incentivise commercial thinking. Needs legal review before implementing.
- Rounding: should base_seat_price round up (ceiling) or nearest cent? Ceiling is slightly safer legally (always slightly below IRS rate).
- Should group bookings (multiple seats in one transaction) apply a single insurance charge or per-rider?
Recommendation: per-rider — each person has independent coverage.
- Annual IRS rate update workflow: admin UI update only, or also push notification to drivers about new pricing on future trips?